Software Project 1 Specification: Stock Data Viewer

Windows Forms Application for Stock Data Analysis

Project Overview

The objective of this assignment is to develop a Windows Forms application in C# that enables users to view and analyze stock data. Students will gain hands-on experience working with file input/output, data visualization, and user interface controls in a desktop environment. The application will read stock candlestick data from CSV files and provide an interactive way for users to filter and review the information.

Requirements

- **Compilation:** Your code MUST compile without errors. If your code doesn't compile, your project will receive a grade of 0.
- **Running:** Obviously, your code must run. A project that doesn't run will receive a grade of 0.
- **Easy Start:** The loading button must have TabIndex 0 so that it has the focus when the application starts. Similarly, the starting and ending date should be January 28, 2021 to February 28, 2021. The OpenFileDialog should also be preset to open the "ABBV-daily". This is so that the grader can hit the "Enter" key twice in a row to see results immediately.
- **Naming Convention:** Make sure you name your controls in the way we specified it in class. You are also to use Camel case, For example the button to load a stock should not be called Button1 (as Visual Studio would) but as Button_SomeMeaningfulName (e.g. Button_LoadStock)
- **File naming:** If you are creating new classes, make sure that the name of the file associated with the new class matches the name of the class. This makes it easy to find where a class is defined/implemented.
- **Comments:** Make sure your code is well commented. Functions headers should be preceded with a "///" comment so that the function's usage is automatically displayed. Inside the function implementation should have "//" type comments inside the function to explain the working of the function.

Application Features

1. **Stock Selection:** The application must display a list of available CSV files from the *Stock Data* folder. Users should be able to select a file representing a specific stock.
2. **Data Display:** Upon selection, the application should read the chosen CSV file and present its contents in a *DataGridView* control.
3. **Date Range Input:** Users must be able to specify a start date and end date to view only the stock data within that range.
4. **Update Functionality:** An *Update* button must be provided. When pressed, the DataGridView should refresh to show data according to the selected date range.
5. **Date Range Modification:** Users should have the option to change the date range and update the display as often as needed.
6. **Period Specification:** The user must be able to display all the stock files or just the daily, just the weekly, monthly or yearly periods.

Technical Specifications

- **Programming Language:** Cpp/CLI (.NET Framework or .NET Core)
- **UI Framework:** Windows Forms
- **Controls:** utilize the controls we used in class.
- **Data Handling:** Use appropriate objects from the .NET framework libraries to read and parse CSV files. DO NOT USE outside classes like CSVHelper. You are to write your own data loading class (as we did in class).

Submission Instructions

- Submit the complete Visual Studio solution by CLEANING (Clean in the menu) and then ZIPPING your solution folder (NOT the project folder which is inside the solution folder). Do not include the “Stock Data” file. That file should be in the parent folder of your solution folder.
- A rubric will be provided shortly.
- I have tried to keep instructions simple to minimize confusion. But feel free to ask questions if you have any.